The Bidirectional Movement of the Frontalis Muscle: Introducing the Line of Convergence and Its Potential Clinical Relevance

Sebastian Cotofana, M.D., Ph.D.
David L. Freytag
Konstantin Frank, M.D.
Sonja Sattler, M.D.
Marina Landau, M.D.
Tatjana Pavicic, M.D.
Sabrina Fabi, M.D.
Nirusha Lachman, Ph.D.
Claudia A. Hernandez, M.D.
Jeremy B. Green, M.D.

Albany, N.Y.; Munich and Darmstadt, Germany; Holon, Israel; San Diego, Calif.; Rochester, Minn.; Medellin, Colombia; and Coral Gables, Fla.

Background: Cosmetic treatment of the forehead using neuromodulators is challenging. To avoid adverse events, the underlying anatomy has to be understood and thoughtfully targeted. Clinical observations indicate that eyebrow ptosis can be avoided if neuromodulators are injected in the upper forehead, despite the frontalis muscle being the primary elevator.

Methods: Twenty-seven healthy volunteers (11 men and 16 women) with a mean age of 37.5 ± 13.7 years (range, 22 to 73 years) and of diverse ethnicity (14 Caucasians, four African Americans, three Asians, and six of Middle Eastern descent) were enrolled. Skin displacement vector analyses were conducted on maximal frontalis muscle contraction to calculate magnitude and direction of forehead skin movement.

Results: In 100 percent of investigated volunteers, a bidirectional movement of the forehead skin was observed: the skin of the lower forehead moved cranially, whereas the skin of the upper forehead moved caudally. Both movements converged at a horizontal forehead line termed the line of convergence, or C-line. The position of the C-line relative to the total height of the forehead was 60.9 ± 10.2 percent in men and 60.6 ± 9.6 percent in women ($p = 0.941$). Independent of sex, the C-line was located at the second horizontal forehead line when counting from superior to inferior (men, $n = 2$; women, $n = 2$). No difference across ethnicities was detected.

Conclusions: The identification of the C-line may potentially guide practitioners toward more predictable outcomes for forehead neuromodulator injections. Injections above the C-line could mitigate the risk of neuromodulator-induced brow ptosis. (*Plast. Reconstr. Surg.* 145: 1155, 2020.)

Downloaded from http://journals.lww.com/plasreconsurg by BhDMf5ePHKbH4TTImqenVL8iJ7OdHJi5nUa8RKXknrhpZw6yOqGFbAGrzZ5rI+4BNlNMNaW0ek= on 04/27/2020

Treating the signs of facial aging has become widely accepted in today's society. Besides soft-tissue filler injections, the administration of facial neuromodulators (i.e., various types of botulinum toxin) to treat facial wrinkles has increased in demand (increase by 845 percent between the years 2000 and 2018 according to the American Society of Plastic Surgeons).[1] Despite the reduced frequency and severity of adverse events when treating the upper face with neuromodulators, the occurrence of eyebrow ptosis and upper eyelid ptosis is undesired and can result in patient dissatisfaction.[2]

The position of the eyebrow needs to be respected, in addition to the shape and the distinctness of horizontal forehead lines (all of those depending on the underlying muscular anatomy). The position of the eyebrow is at a delicate balance between depressors and elevators. Whereas the frontalis muscle is the only elevator, a group of muscles act as depressors: orbicularis oculi, depressor supercilii, corrugator supercilii, and

From the Division of Anatomy, Department of Medical Education, Albany Medical College; the Division of Plastic Surgery, Department of Surgery, Albany Medical Centre; the Department for Hand, Plastic and Aesthetic Surgery, Ludwig Maximilian University of Munich; Rosenpark Klinik; Wolfson Medical Center, Dermatology; private practice; Cosmetic Laser Dermatology; Mayo Clinic College of Medicine and Science, Mayo Clinic; and Skin Associates of South Florida. Dr. Sattler, for the International Society of Dermatologic Surgery.
Received for publication July 15, 2019; accepted November 14, 2019.
Copyright © 2020 by the American Society of Plastic Surgeons
DOI: 10.1097/PRS.0000000000006756

Disclosure: *None of the authors has any commercial associations or financial disclosures that might pose or create a conflict of interest with the methods applied or the results presented in this article. This study received no financial support or funding.*

Copyright © 2020 American Society of Plastic Surgeons. Unauthorized reproduction of this article is prohibited.

procerus muscles. In addition, the morphology of the frontalis muscle and its muscle fascicle angle influence the shape of horizontal forehead lines: a greater muscle fascicle angle that is oriented laterally results in wavy horizontal forehead lines, whereas a smaller and vertically oriented muscle fascicle angle results in straight horizontal forehead lines. Although a helpful guide for less experienced injectors, neuromodulator injection grid maps providing a "recipe" for treating all patients should be avoided.[3–8] Customized treatments respecting individual anatomical variation should instead be used.

The frontalis muscle is considered an eyebrow elevator, and the application of neuromodulators to treat horizontal forehead lines should generally lead to eyebrow ptosis. However, this is inconsistently observed. It is recommended that neurotoxins be applied to the upper forehead instead of the lower forehead to decrease the risk of eyebrow ptosis during treatment.[2] To date, this clinical phenomenon is poorly understood.

The scope of the present analysis was to investigate motion patterns of the skin of the forehead in healthy volunteers during forehead frowning to understand why the application of neuromodulators in the forehead not generally leads to eyebrow ptosis. It is hoped that skin motion patterns can be identified by using vector displacement analyses, which could explain why lower forehead toxin injections carry a greater risk for eyebrow ptosis compared with the upper forehead.

PATIENTS AND METHODS

Study Sample

The sample investigated in the present analysis is a subsample of a study population described previously by Frank et al.[9] The current sample includes 27 healthy volunteers (11 men and 16 women) with a mean age of 37.5 ± 13.7 years (range, 22 to 73 years) and with the following ethnic distribution: 14 Caucasians, four African Americans, three Asians, and six of Middle Eastern descent. Volunteers were excluded from the analysis if previous treatments, operations, or trauma affected the shape and/or the function of the frontalis muscle. In addition, volunteers were excluded from the study if no distinct hairline was present to determine the upper boundary of the forehead. Participants were informed about the aims and procedures of the study. Each participant provided written informed consent for the use of both their data and their associated images. The study was approved by the Institutional Review Board of Ludwig Maximilian University of Munich (protocol number 266-13). This study was conducted in accordance with regional laws (of Germany) and good clinical practice.

Imaging

Three-dimensional images of the full faces of the volunteers were obtained using a Vectra H1 camera system (Canfield Scientific, Inc., Fairfield, N.J.). Baseline and follow-up images were obtained at rest and on maximal forehead frowning. The follow-up image of each participant was automatically aligned to its respective baseline image. Differences in skin position (maximal frowning versus at rest) were computed using skin vector displacement analysis by means of the automated algorithm of the Vectra Software Suite (Figs. 1 through 3).[9] A maximal mean surface deviation of 0.1 mm was tolerated. All measurements were conducted by the same investigator (D.L.F.) to ensure consistency throughout the analyses.

Statistical Analysis

The orientation and the magnitude of forehead skin displacement were recorded and analyzed. Values of skin compression (i.e., reduction in length in percent) and skin displacement (i.e., change in skin position on contraction in millimeters) were calculated based on skin movement. Normal data sample distribution for forehead length, distance of the C-line from the superior orbital rim (absolute and as a percentage), skin compression, and skin displacement were tested by Shapiro-Wilk test and confirmed by means of normality plots and histograms. Differences in values for sexes were calculated using independent sample *t* testing, and results were verified for consistency using the Mann-Whitney *U* test and the bootstrapping method[10,11] because of the small sample size. Differences in values for ethnic subgroups were calculated using analysis of variance testing, and results were verified for consistency using Kruskal-Wallis test and the bootstrapping method. Nonparametric tests were used for calculations on horizontal forehead lines with presentation of the median value and the respective data range (i.e., smallest and greatest values). Differences in skin compression and skin displacement between the region above versus below the C-line were calculated using paired *t* testing and verified by means of the Mann-Whitney *U* test and the bootstrapping method. All statistical analyses were performed using IBM SPSS Version 23 (IBM

Copyright © 2020 American Society of Plastic Surgeons. Unauthorized reproduction of this article is prohibited.

Corp., Armonk, N.Y.) and results were considered statistically significant at a level of $p \leq 0.05$ to guide conclusions.

RESULTS

Forehead Length

The mean length of the forehead (i.e., the distance between the upper margin of the eyebrows and the hairline) was 65.0 ± 8.1 mm in men and 53.4 ± 9.2 mm in women (p = 0.002). No statistically significant difference in mean forehead length between the four analyzed ethnic groups was identified (unstratified for sex): Caucasian, 57.9 ± 10.3 mm; African American, 58.0 ± 5.1 mm; Asian, 65.7 ± 5.6 mm; and Middle Eastern, 55.0 ± 14.6 mm (p = 0.565).

Horizontal Forehead Lines

The median number of observed horizontal forehead lines (irrespective of wavy or straight appearance) was four (range, two to six), which

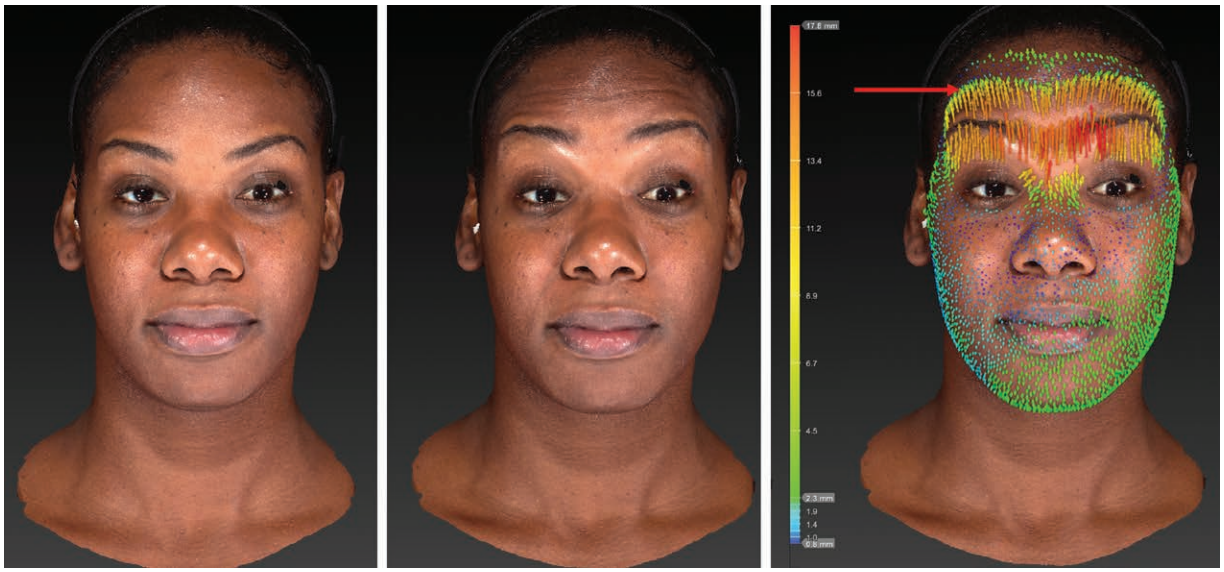


Fig. 1. Processed three-dimensional scan representing the two different states (*left*, resting forehead; *center*, maximal contracted forehead) of the forehead evaluated in this study and their difference in skin displacement represented through vectors (*right*) in a female African American volunteer.

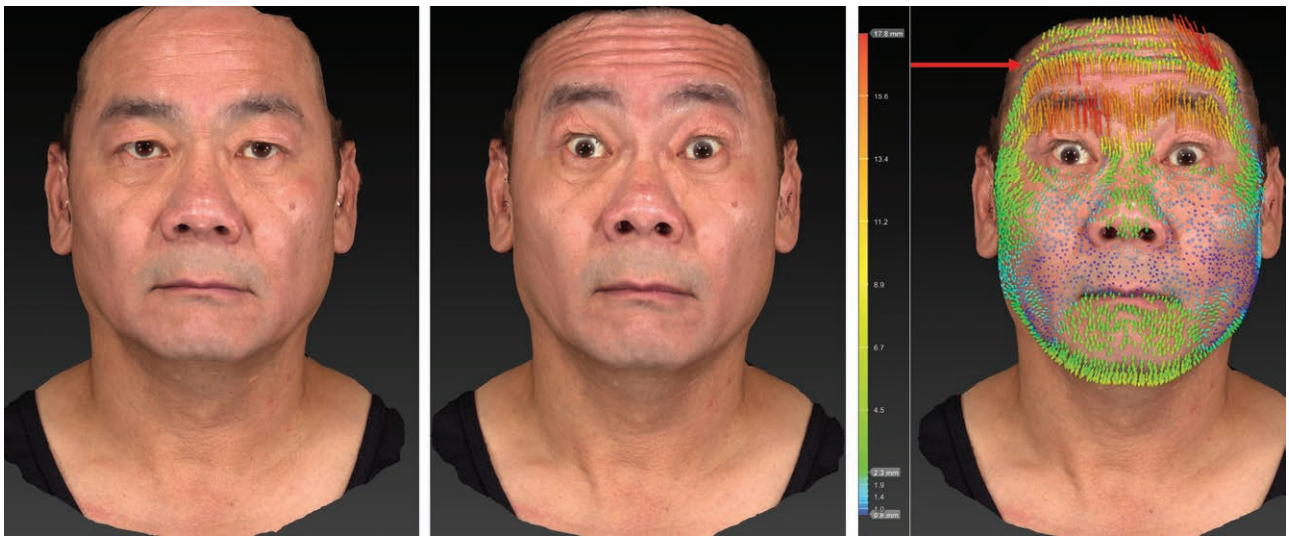


Fig. 2. Processed three-dimensional scan representing the two different states (*left*, resting forehead; *center*, maximal contracted forehead) of the forehead evaluated in this study and their difference in skin displacement represented through vectors (*right*) in a male Asian volunteer.

Copyright © 2020 American Society of Plastic Surgeons. Unauthorized reproduction of this article is prohibited.

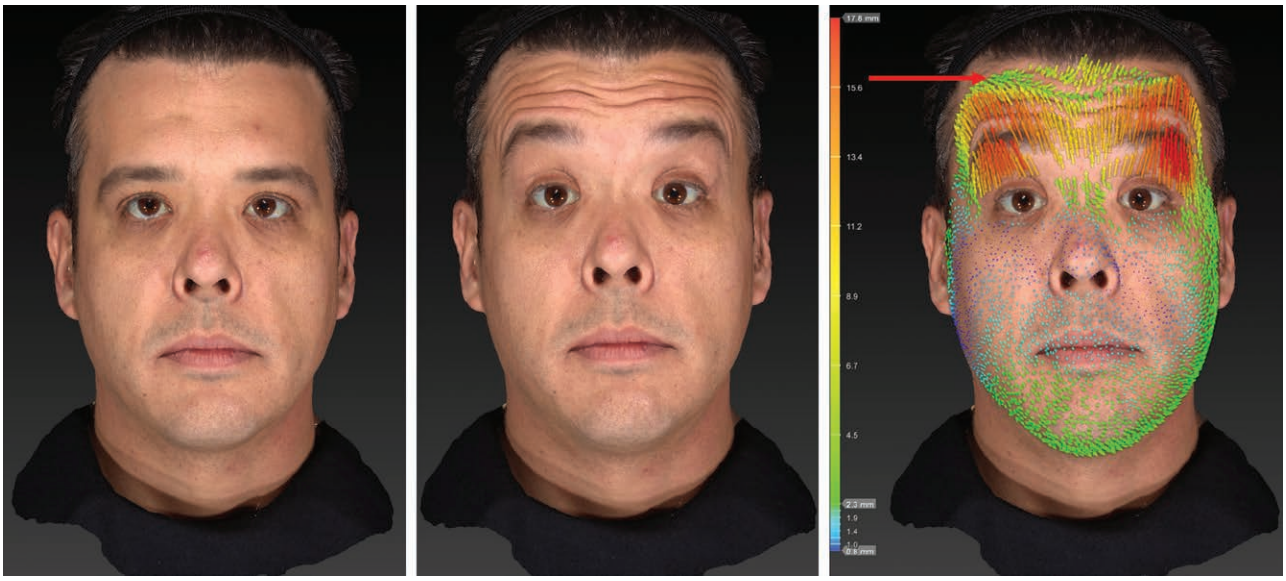


Fig. 3. Processed three-dimensional scan representing the two different states (*left*, resting forehead; *center*, maximal contracted forehead) of the forehead evaluated in this study and their difference in skin displacement represented through vectors (*right*) in a male Caucasian volunteer.

was independent of sex (p = 0.451) or ethnicity (p = 0.148). There was no statistically significant correlation between the number of horizontal forehead lines and the mean length of the forehead (r_p = 0.247; p = 0.214).

Line of Convergence

In all investigated volunteers (100 percent), a bimodal movement of the forehead skin was observed, where the skin of the lower forehead moved cranially and the skin of the upper forehead simultaneously moved caudally. This bimodal movement resulted in an elevation of the eyebrows and a depression of the hairline, and an immobile horizontal forehead line where the two movements converged. This stable horizontal forehead line is termed the line of convergence (i.e., C-line).

The C-line was located above the upper margin of the eyebrows at a mean distance of 39.6 ± 7.7 mm in men and 34.3 ± 7.3 mm in women, with no statistically significant difference between sexes (p = 0.083). Similarly, no significant difference in the location of the C-line was observed across ethnicities (unstratified for sex): Caucasian, 37.4 ± 7.9 mm; African American, 35.4 ± 7.8 mm; Asian, 41.7 ± 11.4 mm; and Middle Eastern, 32.4 ± 5.3 mm (p = 0.370).

The median location of the C-line was at the second horizontal forehead line when counting from the hairline (superior to inferior) irrespective of sex [men, n = 2.0 (range, 2.0 to 3.0); women, n = 2.0 (range, 1.0 to 3.0); p = 1.0]. This was consistent across ethnicities (unstratified for sex): Caucasian, n = 2.0; African American, n = 2.0; Asian, n = 2.0; and Middle Eastern, n = 2.0 (p = 0.731). The median number of horizontal forehead lines below the C-line was 2.0 (range, 0.0 to 4.0) in men and 2.0 (range, 1.0 to 3.0) in women (p = 0.394) and across ethnic groups: Caucasian, n = 1.0; African American, n = 1.0; Asian, n = 2.0; and Middle Eastern, n = 2.0 (p = 0.130).

The position of the C-line relative to the total height of the forehead was 60.9 ± 10.2 percent in men and 60.6 ± 9.6 percent in women (p = 0.941) (Fig. 4). C-line location across ethnicities was (unstratified for sex) as follows: Caucasian, 62.8 ± 8.5 percent; African American, 54.5 ± 10.4 percent; Asian, 61.8 ± 8.6 percent; and Middle Eastern, 59.5 ± 12.6 percent (p = 0.507 across groups).

Difference between Upper (above the C-Line) and Lower (below the C-Line) Forehead Parameters

The compression of skin (i.e., reduction in length) in the lower versus the upper forehead was 26.0 percent versus 22.3 percent (p = 0.089) in men and 20.9 percent versus 13.7 percent (p = 0.001) in women. The gender difference was 5.1 percent (p = 0.039) for the lower forehead and 8.5 percent (p = 0.005) for the upper forehead (Fig. 5).

The total skin displacement (i.e., change in skin position on maximal frontalis contraction)

Copyright © 2020 American Society of Plastic Surgeons. Unauthorized reproduction of this article is prohibited.



Fig. 4. Processed three-dimensional scan showing the location of the C-line at 61 percent of the mean forehead length (from caudal to cranial) (*left*) and at the second horizontal forehead line (*right*).

was 6.8 mm in men for the upper forehead versus 6.1 mm for the lower forehead (p = 0.244). In women, it was 4.8 mm for the upper forehead versus 3.6 mm for the lower forehead (p = 0.020). The gender difference was 2.5 mm (p = 0.004) for the lower forehead and 2.0 mm (p = 0.001) for the upper forehead (Fig. 6).

DISCUSSION

This skin displacement analysis performed in 27 ethnically diverse individuals for the first time quantifies the bidirectional movement of the skin of the forehead on maximal frontalis muscle contraction: the skin of the lower forehead moves cranially, resulting in an elevation of the eyebrows; whereas the skin of the upper forehead moves caudally, resulting in a depression of the hairline. These two movements meet at a horizontally oriented line termed the line of convergence (C-line). The C-line was located 39.6 ± 7.7 mm above the eyebrows in men and 34.3 ± 7.3 mm in women, which corresponds to 61 percent of the total forehead length in both sexes and was independent of ethnicity (Fig. 4). Interestingly, the C-line was located at the second horizontal forehead line inferior to the hairline independent of sex or ethnicity.

A strength of the present study is that 11 men and 16 women with diverse ethnic backgrounds (Caucasian, African American, Asian, and Middle Eastern) were investigated. This study sample might potentially represent the heterogenous patient population seeking medical intervention in current clinical practice. However, the small overall sample size (n = 27) and the small subsamples in each ethnic subgroup (n = 14 Caucasians, n = 4 African Americans, n = 3 Asians, and n = 6 of Middle Eastern descent) should be regarded as a limitation of the study. A larger sample size with greater numbers, especially in the ethnic subsamples, would have been favorable to strengthen the message presented herein. Future studies with larger samples and/or studies carried out in a clinical scenario will need to expand on the concepts described.

Another strength of this study is the methodology used to measure forehead skin displacement. Three-dimensional skin vector displacement analysis has been shown to provide objective quantification of minute skin movements.[9] This noninvasive, real-time evaluation of skin displacement is based on the mathematical computation of the difference in skin position before and after a movement or intervention. The automated algorithm identifies and tracks the movement of prominent two-dimensional skin features such as moles or skin pores and computes a skin displacement vector. Each vector is characterized by a direction and a specific magnitude. These values can be used to locally and regionally compute skin displacement

Copyright © 2020 American Society of Plastic Surgeons. Unauthorized reproduction of this article is prohibited.

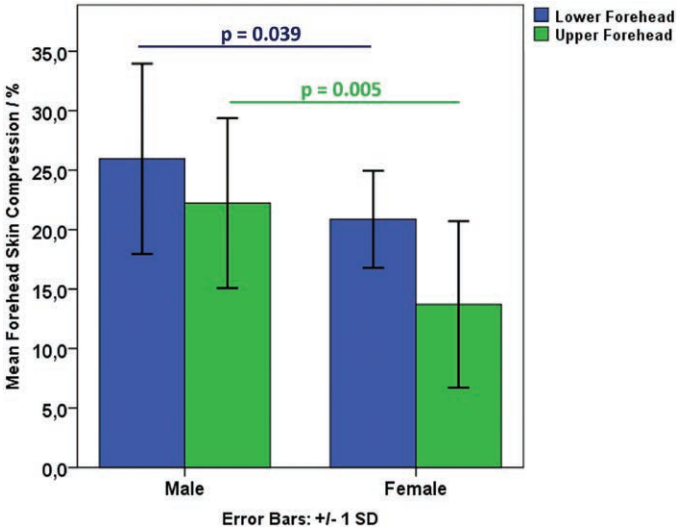


Fig. 5. Bar graph showing the mean forehead skin compression as a percentage for the upper and lower forehead in men and women. Significant differences could be observed between sex for both the upper and the lower forehead ($p = 0.005$ and $p = 0.039$, respectively).

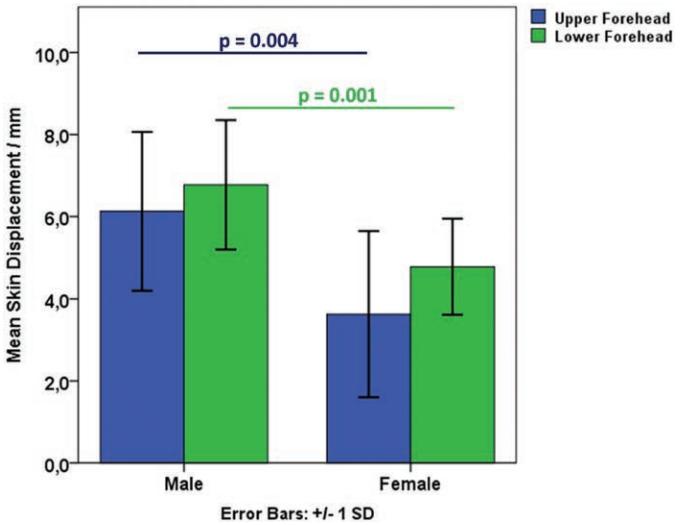


Fig. 6. Bar graph showing the mean skin displacement in millimeters for the upper and lower forehead in men and women. Significant differences could be observed between genders for both the upper and the lower forehead ($p = 0.004$ and $p = 0.001$, respectively).

(change in skin position in millimeters) and skin compression (reduction in length in percent).

In the present analysis, skin vector displacement was used for the total forehead and for the upper versus lower forehead separately. It was found that the skin of the forehead has a bimodal movement. The skin of the lower forehead moved cranially compared with the skin of the upper forehead, which moved caudally. This behavior was observed in 100 percent of the investigated sample. From an anatomical perspective, this is plausible, as the frontalis muscle has no bony origin but is located in its enveloping sheath covered by subcutaneous fat and separated from the deep forehead compartments[12] by the subfrontalis fascia.[13] On contraction of the frontalis muscle, the caudal and the cranial ends are approached, resulting in a bidirectional movement: eyebrow elevation and hairline depression.

Using neuromodulators for the reduction of forehead lines is challenging. The frontalis muscle is the only eyebrow elevator, and its muscle tonus

Copyright © 2020 American Society of Plastic Surgeons. Unauthorized reproduction of this article is prohibited.

has to be balanced against the eyebrow depressors (i.e., orbicularis oculi, depressor supercilii, corrugator supercilii, and procerus muscles). Reducing frontalis muscle contractility and especially the muscle's eyebrow-elevation segment could result in eyebrow ptosis. This adverse event is transient, and its published incidence ranges from 0.6 to 20 percent. Therefore, injectors should be prepared and openly communicate this potential risk to the patient.[14–16]

Although the present study did not evaluate neuromodulator injections a priori, the results of the present study confirm clinical observations. Injections in the lower forehead can cause eyebrow ptosis, whereas injections in the upper forehead may affect forehead lines but not eyebrow position. Previously published concepts[17] and consensus panel recommendations[18] independently mention that injections within a distance of 2.0 to 3.0 cm above the superior orbital rim increase the risk for eyebrow ptosis. No study to date has provided reliable evidence for this clinical observation and why neuromodulator administration in the lower versus the upper forehead can have different effects on eyebrow position. Our study, however, provides for the first-time reliable evidence, as we were able to show that the frontalis muscle has a bimodal contraction pattern. Whereas the lower part of the muscle acts as an eyebrow elevator, the upper part acts as a hairline depressor. This could explain why injections into the lower part of the muscle (i.e., eyebrow-elevation segment) can result in eyebrow ptosis, whereas injections in the upper part of the muscle (i.e., hairline-depressor segment) do not carry this risk. Decreasing the risk of eyebrow ptosis could be potentially achieved by reducing the units of neuromodulators injected below the C-line versus above it or by repositioning injection points more cranially and focusing more on the C-line region and above. The shape of the (untreated) horizontal forehead lines (straight versus wavy lines)[9,19] might also help to individualize neuromodulator treatment and to guide dosage and injection locations.

Of additional interest, skin compression and skin displacement were significantly increased in men compared with women. This can be understood as the result of the greater frontalis muscle activity caused by increased muscle mass resulting in greater contractility and thus in greater skin movement.[20,21] This larger skin displacement attributable to larger muscle mass is why men often require greater than 50 percent higher neuromodulator dosing than women to achieve a comparable clinical effect.

Future investigations could use skin displacement analysis to study the effects on eyebrow position of neuromodulator injections placed above and below the C-line. It would be especially interesting to determine how successful injections above the C-line with neuromodulator are at smoothing forehead rhytides without causing eyebrow ptosis.

CONCLUSIONS

In this study, a bidirectional movement of the skin of the forehead was observed: the skin of the lower forehead moved cranially, whereas the skin of the upper forehead moved caudally. Both motions met at a static, nonmoving line termed the line of convergence (C-line). The C-line was identified at 61 percent of the total mean forehead length, which was consistent in all sexes and ethnicities. Clinically, the C-line can be identified at the second horizontal forehead line when counting from superior to inferior. Placing neuromodulator injections above the C-line may mitigate the risk of eyebrow ptosis.

Sebastian Cotofana, M.D., Ph.D.
Department of Clinical Anatomy
Mayo Clinic College of Medicine and Science
Mayo Clinic
Stabile Building 9-38
200 First Street
Rochester, Minn. 55905
cotofana.sebastian@mayo.edu
Instagram: professorsebastiancotofana
Facebook: professorsebastiancotofana

PATIENT CONSENT

Patients provided written consent for the use of their images.

ACKNOWLEDGMENT

The authors would like to thank Canfield Scientific, Inc., for support with data acquisition and Dr. Fabio M. Ingallina for his guidance and expertise in the initiation phase of the study.

REFERENCES

1. American Society of Plastic Surgeons. 2018 plastic surgery statistics report. Available at: https://www.plasticsurgery.org/documents/News/Statistics/2018/plastic-surgery-statistics-full-report-2018.pdf. Accessed February 24, 2020.
2. King M. Management of ptosis. *J Clin Aesthet Dermatol.* 2016;9:E1–E4.
3. Prager W, Nogueira Teixeira D, Leventhal PS. IncobotulinumtoxinA for aesthetic indications: A systematic review of prospective comparative trials. *Dermatol Surg.* 2017;43:959–966.

Copyright © 2020 American Society of Plastic Surgeons. Unauthorized reproduction of this article is prohibited.

4. Prager W, Wissmüller E, Kollhorst B, Williams S, Zschocke I. Comparison of two botulinum toxin type A preparations for treating crow's feet: A split-face, double-blind, proof-of-concept study. *Dermatol Surg.* 2010;36(Suppl 4):2155–2160.
5. de Maio M, Swift A, Signorini M, Fagien S; Aesthetic Leaders in Facial Aesthetics Consensus Committee. Facial assessment and injection guide for botulinum toxin and injectable hyaluronic acid fillers. *Plast Reconstr Surg.* 2017;140:265e–276e.
6. Kapoor KM, Chatrath V, Anand C, et al. Consensus recommendations for treatment strategies in Indians using botulinum toxin and hyaluronic acid fillers. *Plast Reconstr Surg Glob Open* 2017;5:e1574.
7. Carruthers J, Fagien S, Matarasso SL; Botox Consensus Group. Consensus recommendations on the use of botulinum toxin type a in facial aesthetics. *Plast Reconstr Surg.* 2004;114(Suppl):1S–22S.
8. Ahn BK, Kim YS, Kim HJ, Rho NK, Kim HS. Consensus recommendations on the aesthetic usage of botulinum toxin type A in Asians. *Dermatol Surg.* 2013;39:1843–1860.
9. Frank K, Freytag DL, Schenck TL, et al. Relationship between forehead motion and the shape of forehead lines: A 3D skin displacement vector analysis. *J Cosmet Dermatol.* E-published ahead of print July 8, 2019.
10. Davison AC, Hinkley DV. *Bootstrap Methods and Their Application.* Cambridge: Cambridge University Press; 1997.
11. Chernick MR, LaBudde RA. *An Introduction to Bootstrap Methods with Applications to R.* Hoboken, NJ: Wiley; 2011.
12. Cotofana S, Mian A, Sykes JM, et al. An update on the anatomy of the forehead compartments. *Plast Reconstr Surg.* 2017;139:864e–872e.
13. Cotofana S, Lachman N. Anatomy of the facial fat compartments and their relevance in aesthetic surgery. *J Dtsch Dermatol Ges.* 2019;17:399–413.
14. Lowe NJ, Lowe P. Botulinum toxins for facial lines: A concise review. *Dermatol Ther (Heidelb.)* 2012;2:14.
15. Lorenc ZP, Smith S, Nestor M, Nelson D, Moradi A. Understanding the functional anatomy of the frontalis and glabellar complex for optimal aesthetic botulinum toxin type A therapy. *Aesthetic Plast Surg.* 2013;37:975–983.
16. U.S. Food and Drug Administration. BOTOX indications approved FDA. Available at: https://www.accessdata.fda.gov/drugsatfda_docs/label/2011/103000s5232lbl.pdf. Accessed February 24, 2020.
17. Monheit G. Neurotoxins: Current concepts in cosmetic use on the face and neck: Upper face (glabella, forehead, and crow's feet). *Plast Reconstr Surg.* 2015;136(Suppl):72S–75S.
18. Anido J, Arenas D, Arruabarrena C, et al. Tailored botulinum toxin type A injections in aesthetic medicine: Consensus panel recommendations for treating the forehead based on individual facial anatomy and muscle tone. *Clin Cosmet Investig Dermatol.* 2017;10:413–421.
19. Moqadam M, Frank K, Handayan C, et al. Understanding the shape of forehead lines. *J Drugs Dermatol.* 2017;16:471–477.
20. Janssen I, Heymsfield SB, Wang ZM, Ross R. Skeletal muscle mass and distribution in 468 men and women aged 18-88 yr. *J Appl Physiol (1985)* 2000;89:81–88.
21. Weeden JC, Trotman CA, Faraway JJ. Three-dimensional analysis of facial movement in normal adults: Influence of sex and facial shape. *Angle Orthod.* 2001;71:132–140.

Copyright © 2020 American Society of Plastic Surgeons. Unauthorized reproduction of this article is prohibited.